

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Drive architecture for agentic AI initiatives: designed a production multi-agent system (LangGraph, MCP tool-calling, OpenAI Agents SDK) with planning, tool-use, and memory layers, powering RAG-based decision support at 99%+ uptime.
- Built RAG and grounding pipelines and an LLM evaluation/gateway layer enforcing automated compliance and drift checks for AI governance, blocking 15% of releases that failed validation.
- Architected AWS and GCP infrastructure (Lambda, S3, DynamoDB, Cloud Run, SageMaker) on Kubernetes for real-time agentic inference and distributed model serving at sub-200ms latency.
- Built event-driven pipelines on Kafka for asynchronous agent-to-agent messaging and downstream telemetry, decoupling inference workloads from user-facing request paths.

QureAI (AI Startup)

Jan 2024 – May 2025

ML Infrastructure Engineer

- Diagnosed a user-facing RAG retrieval gap directly from real usage signals, then owned the fix end-to-end: redesigned chunking and prompt strategy, lifting recall 61% to 89% and cutting the underlying complaint pattern from 22% to 6% of sessions.
- Led migration of synchronous PyTorch-backed model serving to async vLLM with autoscaling as an early engineer at this AI startup, cutting p99 inference latency 420ms to 140ms across 50K+ daily requests.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production Java microservices in financial-services distributed systems for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly financial data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Led standardization of failure-handling contracts across 25+ services in partnership with cross-functional teams through code reviews and design write-ups, reducing production escalations 60% in one quarter.

Projects

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports.

LangFetch | tinyurl.com/LangFetch

- LangGraph agent with Pinecone vector retrieval and hybrid re-ranking decomposes multi-table questions into dependency-ordered SQL subqueries; deployable as MCP-compatible server for enterprise data queries.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, TypeScript, Java, JavaScript, SQL

GenAI & Agentic AI: RAG, LLM Orchestration, Planning/Tool-Use/Memory, LangGraph, LangChain, OpenAI Agents SDK, MCP, Vector Databases (Pinecone, FAISS), PyTorch, vLLM

Distributed Systems & Backend: Node.js, FastAPI, REST APIs, Kafka, Distributed Systems, Data Pipelines

Cloud & Infra: AWS (Lambda, EC2, S3, RDS, SQS, SageMaker), GCP (Cloud Run), Kubernetes, Docker, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer